

**PRAGATI ENGINEERING COLLEGE: SURAMPALEM  
(AUTONOMOUS)**

**III B.Tech I Semester Regular Examinations, November - 2025**

**COMPUTER NETWORKS**

**(Common to CSE, CSE (AI&ML), CSE (AI), CSE (DS) and IT)**

**Time: 3 hours**

**Max. Marks: 70**

**Note:**

- i. Question No. 1 shall contain 10 compulsory short answer questions (2 questions from each unit) for a total of 20 marks such that each question carries 2 marks.
- ii. In each of the questions from 2 to the last question, there shall be either/or type questions of 10 marks each.

| Q. No.           | Questions                                                                                                                                                                          | BTL | CO  | Marks |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-------|
| 1.               | a) List any two comparisons between TCP/IP and OSI models.                                                                                                                         | K2  | CO1 | 2M    |
|                  | b) How does a MAN differ from WAN?                                                                                                                                                 | K2  | CO1 | 2M    |
|                  | c) List any one advantage of selective repeat protocol.                                                                                                                            | K2  | CO  | 2M    |
|                  | d) Implement bit stuffing for the bit stream 1001110111111.                                                                                                                        | K2  | CO  | 2M    |
|                  | e) What is the principle of CSMA?                                                                                                                                                  | K1  | CO  | 2M    |
|                  | f) What is a token in token passing mechanism?                                                                                                                                     | K1  | CO  | 2M    |
|                  | g) What is subnet mask?                                                                                                                                                            | K1  | CO  | 2M    |
|                  | h) What is tunneling?                                                                                                                                                              | K2  | CO  | 2M    |
|                  | i) List any two congestion control mechanisms in TCP.                                                                                                                              | K1  | CO  | 2M    |
|                  | j) What is the difference between a server and a client?                                                                                                                           | K2  | CO  | 2M    |
| <b>UNIT – I</b>  |                                                                                                                                                                                    |     |     |       |
| 2.               | a) Explain in detail the functionalities of each layer in OSI reference model.                                                                                                     | K2  | CO1 | 5M    |
|                  | b) Explain the structure of fiber optic cable and describe the functionality of its data transmission.                                                                             | K2  | CO1 | 5M    |
| <b>OR</b>        |                                                                                                                                                                                    |     |     |       |
| 3.               | a) Illustrate the protocol stack of TCP/IP model and specify the use of each of those protocols.                                                                                   | K2  | CO1 | 5M    |
|                  | b) Compare various unguided media.                                                                                                                                                 | K2  | CO1 | 5M    |
| <b>UNIT – II</b> |                                                                                                                                                                                    |     |     |       |
| 4.               | a) A bit stream 10011101 is transmitted using the standard CRC method described in the text. The generator polynomial is 1001. Show the actual bit string transmitted.             | K3  | CO2 | 5M    |
|                  | b) Explain the frame formats of HDLC.                                                                                                                                              | K2  | CO2 | 5M    |
| <b>OR</b>        |                                                                                                                                                                                    |     |     |       |
| 5.               | a) When bit stuffing is used, is it possible for the loss, insertion, or modification of a single bit to cause an error not detected by the checksum? If not, why not? If so, how? | K3  | CO2 | 5M    |
|                  | b) Briefly explain stop and wait protocol.                                                                                                                                         | K2  | CO2 | 5M    |

## UNIT – III

|    |    |                                                                                                                                              |    |     |    |
|----|----|----------------------------------------------------------------------------------------------------------------------------------------------|----|-----|----|
| 6. | a) | Three stations are numbered A, B & C, are contending for the use of a shared channel. Which mechanism does CSMA use to handle the situation? | K3 | CO3 | 5M |
|    | b) | How does TDMA mechanism solve channel allocation problem. Explain.                                                                           | K2 | CO3 | 5M |

## OR

|    |    |                                                                                                                                             |    |     |    |
|----|----|---------------------------------------------------------------------------------------------------------------------------------------------|----|-----|----|
| 7. | a) | What is ALOHA? Write various types of ALOHA? Apply $G = \frac{1}{2}$ , $G=1$ for Slotted ALOHA and find out the effect of it on throughput. | K3 | CO3 | 5M |
|    | b) | Explain about Ethernet protocol.                                                                                                            | K2 | CO3 | 5M |

## UNIT – IV

|    |    |                                                                                                                                                                                                                                                                                                                                        |    |     |    |
|----|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----|----|
| 8. | a) | Give two example computer applications for which connection-oriented service is appropriate. Give two examples for which connectionless service is best.                                                                                                                                                                               | K2 | CO4 | 5M |
|    | b) | An IP datagram has arrived with the following partial information in the header (in hexadecimal):<br>45000054 00030000 2006...<br>i) What is the header size?<br>ii) Are there any options in the packet?<br>iii) What is the size of the data?<br>iv) Is the packet fragmented?<br>v) How many more routers can the packet travel to? | K3 | CO4 | 5M |

## OR

|    |    |                                                                                                                  |    |     |    |
|----|----|------------------------------------------------------------------------------------------------------------------|----|-----|----|
| 9. | a) | Compare datagram and virtual-circuit subnets.                                                                    | K3 | CO4 | 5M |
|    | b) | What is IPv6? Explain the structure of IPv6 Datagram. Compare and contrast the IPv4 header with the IPv6 header. | K3 | CO4 | 5M |

## UNIT – V

|     |    |                                                                     |    |     |    |
|-----|----|---------------------------------------------------------------------|----|-----|----|
| 10. | a) | Demonstrate how connection management is done in TCP                | K2 | CO5 | 5M |
|     | b) | Explain about Non-persistent versus Persistent Connections in HTTP. | K2 | CO5 | 5M |

## OR

|     |    |                                                                    |    |     |    |
|-----|----|--------------------------------------------------------------------|----|-----|----|
| 11. | a) | What is addressing? Explain addressing concept in transport layer. | K2 | CO5 | 5M |
|     | b) | Illustrate Electronic Mail Architecture in detail.                 | K2 | CO5 | 5M |